Lecture 3

Hamilton's principle and Noether's theorem

Objectives. *Lecture 2 derived Lagrange's equations from Newton's law with ideal constraints. This lecture asks what single principle gives them, and why a symmetry gives a conserved quantity. The answer is that they are the conditions for one number, the* action *of a path, to be stationary. That is the course's first answer to its question: a system is a configuration space with a Lagrangian, and a motion is a curve that makes the action stationary. The principle explains why the equations keep their form in every coordinate system, why the Lagrangian is not unique, what momentum is — the boundary term of the variation — and why every symmetry gives a conserved quantity (Noether). It closes with the energy function and the central force, whose motion is planar because angular momentum is conserved.*

3.1 The action and its first variation

Throughout, $q = (q_1, \dots, q_n)$ ranges over an open set $U \subseteq \mathbb{R}^n$: a patch of generalised coordinates on a configuration space Q (Definition 2.3), or $\mathbb{R}^n$ itself. A *Lagrangian* is a C^2 function $\mathcal{L}(q, \dot{q}, t)$ on $U \times \mathbb{R}^n \times I$, I an open interval of times, in which $\dot{q}$ is an independent variable. Lecture 2 produced one, $\mathcal{L} = T - V$ (Equation (2.8)); it is C^2 when r and V are C^3. A *path* is a C^2 map $q\colon [t_0, t_1] \to U$, and its *action* is

$$S[q] = \int_{t_0}^{t_1} \mathcal{L}\big(q(t), \dot{q}(t), t\big)\, \mathrm{d}t. \tag{3.1}$$

S is a function of a path, so "stationary" is defined through one-parameter families.

Definition 3.1 (Variation, stationary path)**.** A *variation* of a path q is a C^2 map $(s, t) \mapsto q(s, t) \in U$ on $(-\varepsilon, \varepsilon) \times [t_0, t_1]$ with $q(0, t) = q(t)$; its *direction* is $\delta q(t) = \partial_s q(0, t)$. It has *fixed ends* if $q(s, t_0)$ and $q(s, t_1)$ do not depend on s. The path q is *stationary* if $\frac{\mathrm{d}}{\mathrm{d}s} S[q(s, \cdot)]\big|_{s=0} = 0$ for every variation with fixed ends.

For a path q write, for $j = 1, \dots, n$,

$$p_j(t) = \frac{\partial \mathcal{L}}{\partial \dot{q}_j}\big(q(t), \dot{q}(t), t\big), \qquad \mathcal{E}_j(t) = \dot{p}_j(t) - \frac{\partial \mathcal{L}}{\partial q_j}\big(q(t), \dot{q}(t), t\big), \tag{3.2}$$

the *Euler–Lagrange expressions* $\mathcal{E} = (\mathcal{E}_j)$; they are continuous, since $\mathcal{L}$ is C^2 and q is C^2. The equations $\mathcal{E} = 0$ are Equation (2.9).

Lemma 3.2 (First variation). *For every variation of q, with fixed ends or not,*

$$\frac{\mathrm{d}}{\mathrm{d}s} S[q(s,\cdot)]\Big|_{s=0} = -\int_{t_0}^{t_1} \mathcal{E}(t)\cdot \delta q(t)\,\mathrm{d}t + \Big[p(t)\cdot\delta q(t)\Big]_{t_0}^{t_1}. \tag{3.3}$$

Proof. The integrand $\mathcal{L}(q(s,t),\partial_t q(s,t),t)$ has a continuous s-derivative, so we may differentiate under the integral:

$$\frac{\mathrm{d}}{\mathrm{d}s} S\Big|_{s=0} = \int_{t_0}^{t_1} \Big(\frac{\partial \mathcal{L}}{\partial q}\cdot \delta q + p\cdot \partial_s\partial_t q\big|_{s=0}\Big)\mathrm{d}t.$$

Mixed partial derivatives of a C^2 map commute, so $\partial_s\partial_t q|_{s=0} = \dot{\delta q}$, and p is C^1 in t. Integrating $p\cdot\dot{\delta q}$ by parts gives $[p\cdot\delta q]_{t_0}^{t_1} - \int \dot p\cdot\delta q\,\mathrm{d}t$, which is (3.3). ∎

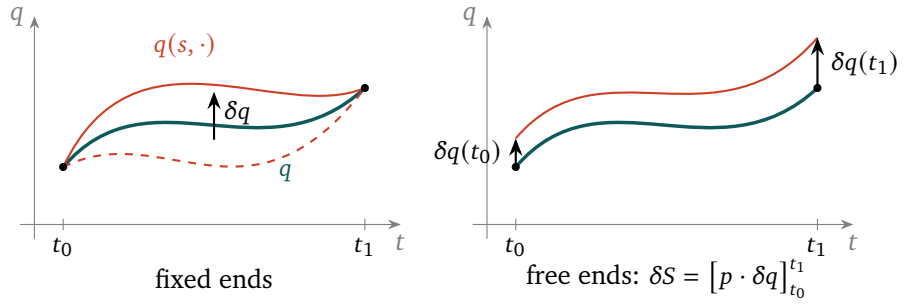


Figure 3.1: Left: a variation with fixed ends (Theorem 3.4). Right: a motion varied with free ends; only the boundary term survives (Proposition 3.10).

Lemma 3.3 (Fundamental lemma of the calculus of variations). *Let $g\colon [t_0,t_1]\to\mathbb{R}^n$ be continuous, and suppose $\int_{t_0}^{t_1} g\cdot\eta\,\mathrm{d}t = 0$ for every C^2 map $\eta\colon[t_0,t_1]\to\mathbb{R}^n$ with $\eta(t_0)=\eta(t_1)=0$. Then $g=0$.*

Proof. If some g_j is not identically 0, then by continuity $g_j(\tau)\neq 0$ at some $\tau\in(t_0,t_1)$, say $g_j(\tau)>0$, and $g_j>0$ on some $[\tau-\varepsilon,\tau+\varepsilon]\subset(t_0,t_1)$. Let η have the single non-zero component $\eta_j=(\varepsilon^2-(t-\tau)^2)^3$ there and 0 outside; its value and first two derivatives vanish at $\tau\pm\varepsilon$, so it is C^2. Then $\int g\cdot\eta\,\mathrm{d}t = \int g_j\eta_j\,\mathrm{d}t > 0$, a contradiction. ∎

Theorem 3.4 (Hamilton's principle). *A path $q\colon[t_0,t_1]\to U$ is stationary if and only if it satisfies the Euler–Lagrange equations*

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial\mathcal{L}}{\partial\dot q_j} - \frac{\partial\mathcal{L}}{\partial q_j} = 0, \qquad j=1,\dots,n, \quad t_0\le t\le t_1. \tag{3.4}$$

Proof. If $\mathcal{E}=0$, the right side of (3.3) vanishes for every variation with fixed ends, since then $\delta q(t_0)=\delta q(t_1)=0$. Conversely, for C^2 η vanishing at the ends, $q+s\eta$ lies in U for $|s|$ small (the image of q is compact, U open): a variation with fixed ends and direction η. So $\int\mathcal{E}\cdot\eta\,\mathrm{d}t=0$ for all such η, and Lemma 3.3 gives $\mathcal{E}=0$. ∎

The principle as the axiom

By Theorem 2.10, for $\mathcal{L} = T - V$ the stationary paths are exactly Newton's motions with ideal constraints; with no constraints (3.4) *is* $m_i\ddot{r}_i = -\partial V/\partial r_i$. From here on the logic is reversed: a system is a configuration space with a Lagrangian, its motions are the stationary paths, and $F = ma$ is a theorem about one Lagrangian. Nothing below uses Newton's law.

3.2 Stationary, not least

The principle is often called the principle of *least* action. It is not one. Take the oscillator, $\mathcal{L} = \frac{1}{2}(\dot{q}^2 - \nu^2 q^2)$ (the frequency is ν, not ω, which the course keeps for the symplectic form of Lecture 6), a motion q_* on $[0, t_1]$, and a competitor $q_* + \eta$ with the same ends, $\eta(0) = \eta(t_1) = 0$. Since $\mathcal{L}$ is quadratic,

$$S[q_* + \eta] = S[q_*] + \int_0^{t_1} (\dot{q}_*\dot{\eta} - \nu^2 q_*\eta)\,\mathrm{d}t + S_2[\eta], \qquad S_2[\eta] = \frac{1}{2}\int_0^{t_1} (\dot{\eta}^2 - \nu^2\eta^2)\,\mathrm{d}t,$$

and the middle term is $[\dot{q}_*\eta]_0^{t_1} = 0$ on a motion. So $S[q_* + \eta] - S[q_*] = S_2[\eta]$ exactly. For $\eta_k = \sin(k\pi t/t_1)$,

$$S_2[\eta_k] = \frac{t_1}{4}\left(\frac{k^2\pi^2}{t_1^2} - \nu^2\right). \tag{3.5}$$

Proposition 3.5 (A saddle, not a minimum). *Let q_* be a motion of the oscillator on $[0, t_1]$ with $t_1 > \pi/\nu$. Then $S[q_* + \varepsilon\eta_1] < S[q_*]$ and $S[q_* + \varepsilon\eta_k] > S[q_*]$ for $\varepsilon \neq 0$ and every $k > \nu t_1/\pi$: q_* is a saddle of the action among paths with its ends.*

Proof. By (3.5), $S[q_* + \varepsilon\eta_k] - S[q_*] = \varepsilon^2 S_2[\eta_k]$, whose coefficient is negative for $k = 1$ when $t_1 > \pi/\nu$ and positive when $k\pi/t_1 > \nu$. ∎

For every t_1 the modes with $k\pi/t_1 > \nu$ raise the action, so a motion is never a maximum; for long times it is a saddle (Figure 3.2). Theorem 3.4 asks only for stationarity, and that is all dynamics needs.

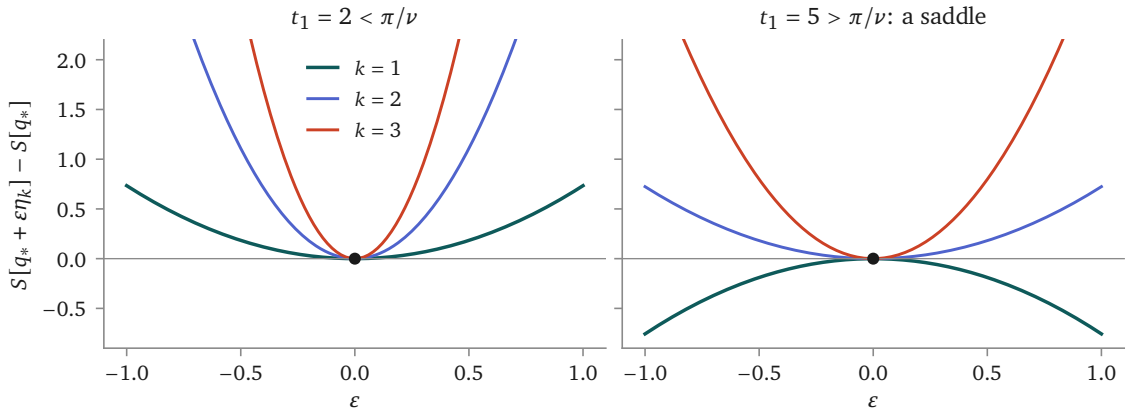


Figure 3.2: The oscillator action along the variations $\varepsilon\eta_k$, $\nu = 1$, which is exactly the parabola $S_2[\eta_k]\varepsilon^2$ of (3.5). For a time $t_1 = 2$, shorter than half a period, every drawn mode raises the action. For $t_1 = 5$ the first mode lowers it and the others raise it: a saddle.

3.3 What the variational form buys

Covariance

Newton's law $m\ddot{x} = F$ changes its form in curvilinear coordinates: in polar coordinates a centrifugal term appears (Example 2.12). The Euler–Lagrange equations do not change their form, and the reason is that the action does not refer to coordinates.

Theorem 3.6 (Covariance under point transformations). *Let $V \subseteq \mathbb{R}^n$ be open and $\Phi\colon V\times I \to U$ be C^3, such that for each t the map $Q \mapsto \Phi(Q,t)$ is a bijection of V onto U with invertible derivative $D_Q\Phi$. Define*

$$\tilde{\mathcal{L}}(Q,\dot{Q},t) = \mathcal{L}\big(\Phi(Q,t),\, D_Q\Phi(Q,t)\,\dot{Q} + \partial_t\Phi(Q,t),\, t\big). \tag{3.6}$$

Then a C^2 path $Q(t)$ satisfies the Euler–Lagrange equations of $\tilde{\mathcal{L}}$ if and only if $q(t) = \Phi(Q(t),t)$ satisfies those of $\mathcal{L}$.

Proof. $\tilde{\mathcal{L}}$ is C^2, because Φ is C^3. The map $(Q,t) \mapsto (\Phi(Q,t),t)$ is a C^3 bijection of $V\times I$ onto $U\times I$ with invertible derivative, so its inverse is C^3 (inverse function theorem). Hence $Q(s,t) \mapsto q(s,t) = \Phi(Q(s,t),t)$ is a bijection between the C^2 variations of Q and those of q, with fixed ends corresponding to fixed ends. The second slot of (3.6) is the chain-rule velocity of $q(s,t)$, so the integrands agree pointwise and

$$\tilde{S}[Q(s,\cdot)] = S[q(s,\cdot)] \qquad \text{for every } s.$$

So Q is stationary for $\tilde{S}$ if and only if q is stationary for S, and Theorem 3.4, applied twice, proves the equivalence. ∎

In particular two patches of Q with a C^3 transition map give the same motions on their overlap. The equations hold at each instant, so a path crossing from patch to patch is a *motion* if it is stationary on every subinterval inside one patch.

So the recipe never changes: substitute, and write (3.4), even for time-dependent transformations.

Gauge freedom

Proposition 3.7 (Equivalent Lagrangians). *Let $F(q,t)$ be C^3 on $U\times I$ and*

$$\mathcal{L}' = \mathcal{L} + \frac{\mathrm{d}F}{\mathrm{d}t}, \qquad \frac{\mathrm{d}F}{\mathrm{d}t}(q,\dot{q},t) := \frac{\partial F}{\partial q}\cdot\dot{q} + \frac{\partial F}{\partial t}.$$

Then $\mathcal{E}' = \mathcal{E}$ along every path: $\mathcal{L}$ and $\mathcal{L}'$ have the same Euler–Lagrange equations.

Proof. Along a path, $S'[q] = S[q] + F(q(t_1),t_1) - F(q(t_0),t_0)$, which does not change under a variation with fixed ends. By (3.3) for both, $\int (\mathcal{E}' - \mathcal{E})\cdot\eta\,\mathrm{d}t = 0$ for every C^2 η vanishing at the ends, and Lemma 3.3 gives $\mathcal{E}' = \mathcal{E}$. ∎

So $\mathcal{L}$ is not determined by the motions.

The charged particle

A particle of mass m and charge e in an electromagnetic field with vector potential $\mathbf{A}(x,t)$ and scalar potential $\phi(x,t)$ (not an angle), both C^3, has

$$\mathcal{L} = \tfrac{1}{2}m\,|v|^2 + e\,v\cdot\mathbf{A}(x,t) - e\,\phi(x,t), \qquad v = \dot{x}, \tag{3.7}$$

with fields $\mathbf{E} = -\nabla\phi - \partial_t\mathbf{A}$ and $\mathbf{B} = \nabla\times\mathbf{A}$.

Proposition 3.8 (Lorentz force). *The Euler–Lagrange equations of* (3.7) *are* $m\ddot{x} = e\big(\mathbf{E} + \dot{x}\times\mathbf{B}\big)$.

Proof. Here $\partial\mathcal{L}/\partial v = mv + e\mathbf{A}$, whose derivative along a path is $m\ddot{x} + e\,\partial_t\mathbf{A} + e(\dot{x}\cdot\nabla)\mathbf{A}$, while $\partial\mathcal{L}/\partial x = e\nabla(v\cdot\mathbf{A}) - e\nabla\phi$ with v held fixed. For fixed v, $\nabla(v\cdot\mathbf{A}) - (v\cdot\nabla)\mathbf{A} = v\times(\nabla\times\mathbf{A})$. So (3.4) reads $m\ddot{x} = e(-\nabla\phi - \partial_t\mathbf{A}) + e\,\dot{x}\times\mathbf{B}$. ∎

If $\mathbf{A}\neq 0$, the term $e\,v\cdot\mathbf{A}$ is linear in the velocity, and $\mathcal{L}$ is not of the form $T - V(x,t)$. The potentials are not unique. For a C^4 function $\chi(x,t)$, the *gauge transformation* $\mathbf{A}' = \mathbf{A} + \nabla\chi$, $\phi' = \phi - \partial_t\chi$ leaves $\mathbf{E}$ and $\mathbf{B}$ unchanged, and changes $\mathcal{L}$ by $e\,\partial_t\chi + e\,v\cdot\nabla\chi = \mathrm{d}(e\chi)/\mathrm{d}t$, as Proposition 3.7 requires.

3.4 The boundary term and momentum

Definition 3.9 (Generalised momentum). The *generalised momentum* conjugate to q_j is $p_j = \partial\mathcal{L}/\partial\dot{q}_j(q,\dot{q},t)$; along a path it is the $p(t)$ of (3.2).

Proposition 3.10 (The boundary term). *If q is a motion, then for every variation of q, with the ends free,*

$$\frac{\mathrm{d}}{\mathrm{d}s}S[q(s,\cdot)]\Big|_{s=0} = p(t_1)\cdot\delta q(t_1) - p(t_0)\cdot\delta q(t_0). \tag{3.8}$$

Proof. $\mathcal{E} = 0$ in (3.3). ∎

On a motion the interior of the path contributes nothing to the first variation; what is left is the momentum at the ends, paired with the displacement of the ends (Figure 3.1, right). This is the definition of momentum that the rest of the course uses. Lecture 8 turns the pairing $p\cdot\delta q$ into the *canonical 1-form* $p\,\mathrm{d}q$.

Examples. For $\mathcal{L} = \tfrac{1}{2}m\,|\dot{x}|^2 - V$, $p = m\dot{x}$. In polar coordinates in the plane (Example 2.12) the momentum conjugate to the angle is $p_\varphi = m\rho^2\dot{\varphi}$, an angular momentum; generalised momenta need not have the units of momentum. For the charged particle $p = mv + e\mathbf{A}$, which is not mv; since $\mathcal{L} + \mathrm{d}F/\mathrm{d}t$ has momentum $p + \partial F/\partial q$, this p changes under a gauge transformation. Momentum belongs to the Lagrangian, not to the motion alone.

Corollary 3.11 (Cyclic coordinates). *If $\partial\mathcal{L}/\partial q_j = 0$ identically, q_j is called* cyclic, *and p_j is constant along every motion.*

Proof. The jth equation (3.4) reads $\dot{p}_j = 0$. ∎

So in polar coordinates the free particle, or any particle in a potential that depends on ρ only, keeps $m\rho^2\dot{\varphi}$. A cyclic coordinate is a symmetry seen in well-chosen coordinates; the next section finds the conserved quantity without choosing them.

3.5 Noether's theorem (Lagrangian)

Definition 3.12 (Quasi-symmetry). A *one-parameter family of point transformations* is a C^3 map $(s,q,t) \mapsto h^s(q,t) \in U$, $|s| < s_0$, with $h^0(q,t) = q$. Its *generator* is $\xi(q,t) = \partial_s h^s(q,t)|_{s=0}$, and it moves a velocity $\dot q$ at (q,t) to $\dot q^s = D_q h^s\,\dot q + \partial_t h^s$, the velocity of $h^s(q(t),t)$. The family is a *quasi-symmetry* of $\mathcal{L}$ if there are functions $F_s(q,t)$, C^2 in (s,q,t), with

$$\mathcal{L}\big(h^s(q,t),\dot q^s,t\big) = \mathcal{L}(q,\dot q,t) + \frac{\partial F_s}{\partial q}\cdot\dot q + \frac{\partial F_s}{\partial t} \tag{3.9}$$

for all $s,q,\dot q,t$, and a *symmetry* if this holds with $F_s = 0$. Write $G(q,t) = \partial_s F_s(q,t)|_{s=0}$.

If each $h^s(\cdot,t)$ is a bijection of U with invertible derivative, Theorem 3.6 and Proposition 3.7 show that a quasi-symmetry maps motions to motions: $h^s(q(t),t)$ is a motion whenever $q(t)$ is. The theorem below does not need this.

Theorem 3.13 (Noether). *If h^s is a quasi-symmetry of $\mathcal{L}$, then along every motion*

$$I = p\cdot\xi(q,t) - G(q,t) \tag{3.10}$$

is constant.

Proof. Differentiate (3.9) in s at $s=0$. Since $h^0 = \mathrm{id}$, $\partial_s \dot q^s|_{s=0} = D_q\xi\,\dot q + \partial_t\xi$, and F_s is C^2, so ∂_s commutes with ∂_q and ∂_t:

$$\frac{\partial\mathcal{L}}{\partial q}\cdot\xi + \frac{\partial\mathcal{L}}{\partial\dot q}\cdot\big(D_q\xi\,\dot q + \partial_t\xi\big) = \frac{\partial G}{\partial q}\cdot\dot q + \frac{\partial G}{\partial t}.$$

Evaluate along a motion $q(t)$. There $\partial\mathcal{L}/\partial q = \dot p$ by (3.4), $D_q\xi\,\dot q + \partial_t\xi = \frac{\mathrm{d}}{\mathrm{d}t}\xi(q(t),t)$, and the right side is $\frac{\mathrm{d}}{\mathrm{d}t}G(q(t),t)$. So the identity reads $\frac{\mathrm{d}}{\mathrm{d}t}\big(p\cdot\xi\big) = \frac{\mathrm{d}}{\mathrm{d}t}G$. ∎

For a symmetry, $I = p\cdot\xi$ is the momentum paired with the direction in which the symmetry moves the configuration (Figure 3.3). Where the translations $h^s q = q + s e_j$ stay in the patch U (for instance when $U = \mathbb{R}^n$), a cyclic q_j is the case $I = p_j$.

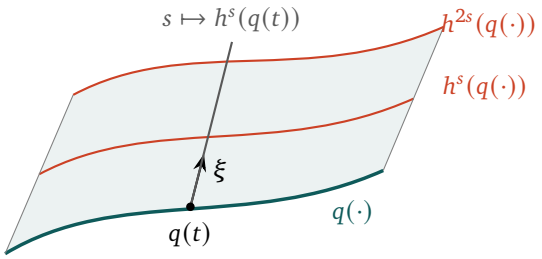


Figure 3.3: Noether's theorem. A quasi-symmetry carries the motion $q(\cdot)$ (bold) to the motions $h^s(q(\cdot))$, which sweep out a surface in configuration space. At each time the curve $s \mapsto h^s(q(t))$ leaves $q(t)$ with velocity ξ, and the conserved quantity is $p\cdot\xi - G$. After Arnold, *Mathematical Methods*, Fig. 70.

Examples

Take N particles in $\mathbb{R}^3$, positions $\mathbf{q}_a$, masses m_a, with $\mathcal{L} = \sum_a \tfrac12 m_a|\dot{\mathbf{q}}_a|^2 - V$, and let $\mathbf{e}$ be a unit vector.

1. *Rotations.* If V is unchanged by rotations about the origin, let h^s rotate every $\mathbf{q}_a$ by the angle s about $\mathbf{e}$. Then $\xi_a = \mathbf{e} \times \mathbf{q}_a$, and $I = \sum_a p_a \cdot (\mathbf{e} \times \mathbf{q}_a) = \mathbf{e} \cdot \mathbf{L}$ with $\mathbf{L} = \sum_a \mathbf{q}_a \times p_a$: *angular momentum* is conserved.
2. *Uniform gravity*, a quasi-symmetry. For $\mathcal{L} = \frac{1}{2}m\dot{z}^2 - mgz$ the shift $z \mapsto z + s$ changes $\mathcal{L}$ by $-mgs = \frac{\mathrm{d}}{\mathrm{d}t}(-mgst)$. So $F_s = -mgst$, $G = -mgt$, and $I = p + mgt$ is constant: $p(t) = p(0) - mgt$. A quasi-symmetry can give a conserved quantity that depends on time.

3.6 The energy function and the central force

A shift of time is not a point transformation of Q, and Theorem 3.13 does not apply to it as stated. The conserved quantity is found directly.

Theorem 3.14 (The energy function). *Let $E_{\mathcal{L}} = p \cdot \dot{q} - \mathcal{L}$. Along every motion*

$$\frac{\mathrm{d}}{\mathrm{d}t} E_{\mathcal{L}} = -\frac{\partial \mathcal{L}}{\partial t}.$$

In particular, if $\mathcal{L}$ does not depend on t, $E_{\mathcal{L}}$ is constant.

Proof. Along a path, $\frac{\mathrm{d}}{\mathrm{d}t}(p \cdot \dot{q}) = \dot{p} \cdot \dot{q} + p \cdot \ddot{q}$ and $\frac{\mathrm{d}}{\mathrm{d}t}\mathcal{L} = \frac{\partial \mathcal{L}}{\partial q} \cdot \dot{q} + p \cdot \ddot{q} + \frac{\partial \mathcal{L}}{\partial t}$. Subtract, and use $\dot{p} = \partial \mathcal{L}/\partial q$. ∎

$E_{\mathcal{L}}$ is the *energy function*. For the Lagrangian $\mathcal{L} = T - V$ of Lecture 2, with constraints that do not move, $T = \frac{1}{2}\dot{q}^{\mathsf{T}} G(q) \dot{q}$ is a quadratic form in $\dot{q}$, so $p \cdot \dot{q} = \dot{q}^{\mathsf{T}} G \dot{q} = 2T$ and

$$E_{\mathcal{L}} = 2T - (T - V) = T + V, \tag{3.11}$$

the energy. Where velocities can be traded for momenta (Theorem 4.5 says when), $E_{\mathcal{L}}$ becomes the Hamiltonian.

The central force

Let $\mathcal{L} = \frac{1}{2}m\,|\dot{\mathbf{q}}|^2 - V(|\mathbf{q}|)$ on $\mathbb{R}^3 \setminus \{0\}$, V of class C^2 on $r > 0$. By the rotation example of § 3.5, $\mathbf{L} = \mathbf{q} \times p$ is conserved: this is Noether's payoff.

Proposition 3.15 (Planarity). *Along a motion:*

1. *if $\mathbf{L} \neq 0$, the motion lies in the plane through 0 perpendicular to $\mathbf{L}$; if $\mathbf{L} = 0$, it lies on a ray from 0;*
2. *in polar coordinates (r, φ) in that plane (any plane containing the ray, if $\mathbf{L} = 0$), φ is cyclic and $p_\varphi = mr^2\dot{\varphi} = \ell$ is constant.*

Proof. (1) $\mathbf{q} \cdot \mathbf{L} = \mathbf{q} \cdot (\mathbf{q} \times p) = 0$. If $\mathbf{L} = 0$, then $\mathbf{q} \times \dot{\mathbf{q}} = 0$ and $\frac{\mathrm{d}}{\mathrm{d}t}\frac{\mathbf{q}}{|\mathbf{q}|} = -\frac{\mathbf{q} \times (\mathbf{q} \times \dot{\mathbf{q}})}{|\mathbf{q}|^3} = 0$. (2) In polar coordinates in the plane, $\mathcal{L} = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\varphi}^2) - V(r)$ (Example 2.12), which does not contain φ; Corollary 3.11 applies. ∎

So the central force is a planar problem with one conserved angular momentum, and $r(t)$ obeys a one-degree-of-freedom equation (the radial Euler–Lagrange equation with $\dot{\varphi} = \ell/mr^2$); Lecture 10 reduces it to the effective potential and solves Kepler's problem.

Summary.

- *The first answer:* a system is a configuration space with a Lagrangian, and its motions are the paths that make the action stationary with fixed ends (Theorem 3.4) — stationary, not least (Proposition 3.5).
- *Handed on:* the equations keep their form under point transformations (Theorem 3.6); adding $\mathrm{d}F/\mathrm{d}t$ to $\mathcal{L}$ leaves them unchanged (Proposition 3.7); the momentum $p = \partial\mathcal{L}/\partial\dot{q}$ is the boundary term of the variation (Proposition 3.10); a quasi-symmetry conserves $p \cdot \xi - G$ (Theorem 3.13); the energy function $E_{\mathcal{L}}$ (Theorem 3.14); the central force is planar (Proposition 3.15).
- *Next:* Lecture 4 trades the velocity for the momentum, and $E_{\mathcal{L}}$ becomes the Hamiltonian.

References

Arnold, *Mathematical Methods of Classical Mechanics*, §§12–13 (pp. 55–61: variations and the Euler–Lagrange equations) and §20 (pp. 88–90: Noether's theorem, whose figure Figure 3.3 follows). Tong, *Lectures on Classical Dynamics*, §2.1 (pp. 10–12: "a minimum or a saddle point"), §2.4 (pp. 23–26: Noether) and §2.5.7 (pp. 36–37: the charged particle and gauge freedom).